

Q1

Select the correct answer from multiple choices given to you on question paper.

1- A field in a relation is also called an_____

- a) Record b) Tuple c) Attribute d) All of the above

2- A record in database is called a _____

- a) RDBMS b) Database c) Tuple d) Relation

3- Referential integrity constraint defines the relationship between the Primary and the_____key.

- a) Alternate b) Foreign Key c) Composite Primary d) Candidate Key

4- Primary key attribute should not contain_____ values

- a) Duplicate b) Unique c) Single d) Double key .

5- Candidate key which is not chosen as the primary key is known as_____key.

- a) Primary Key b) Candidate Key c) Foreign Key d) Attribute key

6- The first stage in the design of a database is to define the_____ view

- a) External b) Conceptual c) Internal d) Physical

7- _____statement is used to retrieve information from a database table .

- a) Retrieve b) From c) Where d) Select

8- _____defines whether all or only some entity occurrences participate in a relationship .

- a) participation b) Primary c) Cardinality d) None of the above

9- SQL function UNIQUE (Q) returns TRUE if there are no_____ tuples in the result of a query .

- a) Duplicate b) Single c) Unique d) All of a , b & c

10- Built in aggregate functions in SQL are_____

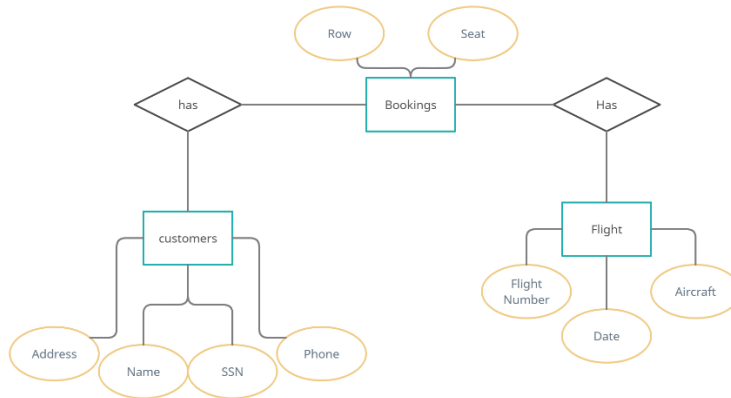
- a) COUNT b) SUM c) AVG , MIN & MAX d) All of a , b & c



Q2

a) In the following diagram a segment of an Entity Relationship Diagram is given. Make the Relational Schema Design for the given ER Design.

b) You have to specify the **Relations, Attributes, Primary and Foreign Keys and Relationships between the tables.**



Customers			
<u>SSN</u>	Name	Phone	

Bookings				
<u>SSN</u>	<u>Flight Number</u>	<u>Date</u>	Row	Seat

Flight		
<u>Flight Number</u>	<u>Date</u>	Aircraft

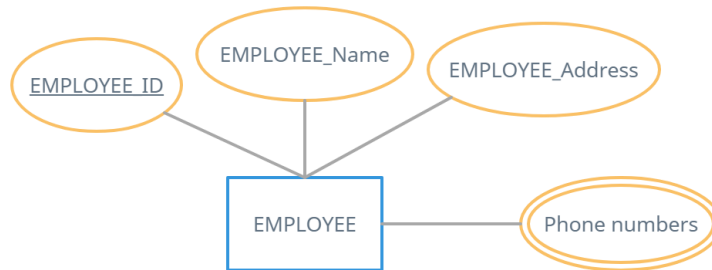
Q3

Write the names of the three main types of the Relational Database Constraints. Also explain those Constraints in one line.



Q4

In the following given ER segment, how a multi-value attribute will be transformed into a relational schema. Draw the relational schema for the given ER segment. Draw the relationship between the tables as well. Underline the Primary and Foreign Keys in the relations.



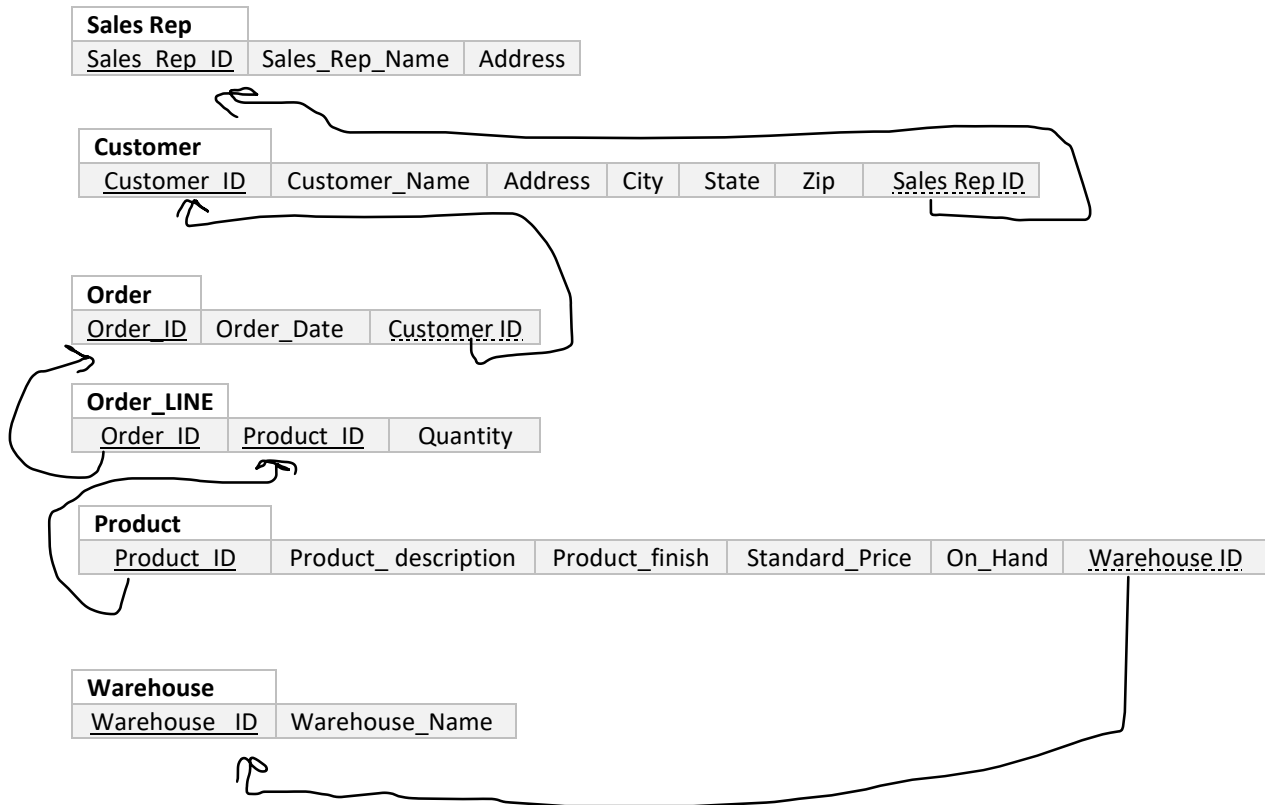
Q5

What is the difference between Primary Key, Composite Primary Key and Candidate Key? Write your answer in one to two lines only.



Q6

Complete the following Relational Database Schema Design by identifying the Primary and the Foreign Keys and by making the Relationships between the referencing and the referenced relations on the below given relational schema diagram.



Q7

Complete the following table by writing the informal terms for the given formal terms used in Relational Databases .

فالسؤال رح يعطيك جدول ال Formal ويطلب منك تكتبها بال Informal

<u>Informal Terms</u>		<u>Formal Terms</u>
Table		Relation
Column Header		Attribute
All possible Column Values		Domain
Row		Tuple, Record or Instance
Table Definition		Schema of a Relation
Populated Table		State of the Relation

Q8

Write down the SQL queries based on the following two statements:

(i) For every project located in " Stafford ", list the project number, the controlling department number, and the department manager's last name, address, and birth date.

```
SELECT PNUMBER, DNUM, LNAME, ADDRESS, BDATE
FROM PROJECT, DEPARTMENT, EMPLOYEE
WHERE DNUM = DNUMBER AND MGRSSN = SSN AND PLOCATION = 'Stafford';
```

(ii) Retrieve all the employees in department 5 whose salary is between \$ 30,000 and \$ 40,000.

```
SELECT *
FROM EMPLOYEE
WHERE (SALARY BETWEEN 30000 AND 40000) AND DNO = 5;
```

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